UNIT 15 MICROFILIMING AND DIGITISATION

Structure

15.0 OBJECTIVES

After reading this Unit, you will be able to:

- understand the basic concept of microfilming, the various microforms, types of microfilm;
- discuss the various advantages and disadvantages of microfilm as a medium for preservation;
- understand the digitisation and its basic concepts;
- learn about the digitisation of text, images, audio and video;
- discuss the various digitisation projects going on all over the world and in India;
- discuss the advantages and disadvantages of digitisation.

15.1 INTRODUCTION

Newspapers, books, manuscripts and archives have been filmed for decades in order to protect them from the deterioration of paper, or from other causes of damage, which threaten books and archival material, and to ensure the permanence of the information they contain. Microfilm is used in libraries for storing back issues of magazines/newspapers and in museums for storing old valuable documents so that the originals need not be handled. Large organisations may also store documents on microfilm to save storage space. When properly produced and stored in accordance with international standards, microfilm has the advantage of maintaining access to information for hundreds of years. Digitisation has introduced possibilities that seem limitless and advantages like enhanced access, diminished costs, versatile capabilities and usage. Libraries and institutions are now taking initiatives to exploit the new media for preservation purposes However, the technology for preserving digital materials is still in developmental stage and requires substantial investments to be made. This unit discusses some of the issues related to microfilming and digitisation and their significance in the context of preservation.

15.2 MICROFILMING

Preservation of a document often involves copying or reformatting of information from one form to another in order to preserve it. This is also known as preservation reformatting. Microfilming is an example of reformatting technique. Other techniques are photocopying and digitisation.

In microfilming images are reduced to such a small size that they cannot be read without optical assistance. This photographic compression leads to saving of space and is of enduring value. Microfilm has an estimated lifespan of 500+ years when stored in proper condition. Moreover, a roll of 35mm microfilm can hold about 900 pages and a roll of 16mm about 3000 pages.

Rare valuable archival documents deteriorate with time. This could be because the paper they are recorded on is of poor quality, the conditions of storage are adverse or because of frequent use. By microfilming the information recorded in these documents can be preserved and retrieved/used as and when required. This saves the original document from further damage even after the original has deteriorated and become unusable. The original documents or records could be in the form of brittle and decaying books, newspapers, maps, plans, and archival documents such as diaries and manuscripts. These rare and valuable records can be microfilmed to preserve them from loss and destruction over time. In the case of valuable documents, which might become damaged by constant use, a microfilm copy of it may be made and stored separately. The film used is safety film and if properly processed it will last longer than the originals. If possible, the microfilm copy may be given to the reader for reference purposes which will not only prevent the original from damage by constant use but will also protect it from danger, such as fire, natural disaster, etc.

15.2.1 History and Concept

Although the principles of microfilming have been known for over 150 years, it is only after the Second World War that the use of microfilming methods became very popular as a technique for reproducing the printed page. Soon after the invention of photography, the need and applications of reduced size of copies were realized. In 1839, John Benjamin Dancer, an English optician, produced micro-documents, which could be viewed with a microscope. He reduced a 20” documents to an image of 1/8” in length. Utilising Dancer’s techniques a French photographer and chemist, Rene Dagron, obtained the first patent for microfilm in 1859 and began the first commercial microfilming enterprise. The application of this technique came into prominence only during the Franco-Prussian war in 1870 when carrier pigeons were used to transport microfilmed messages across German lines to the besieged city of Paris. During World War II microphotography was used extensively for espionage and for regular military mail. Letters going overseas were sent on microfilm, and a hardcopy was produced and forwarded at the receiving side. Because of war a threat of destruction to the records of civilization was realised, which led to the need for the microfilming of records, documents, archives and collections. After the war it was proposed that microforms be used for active information systems and for preservation of material. In the 1970’s the information explosion forced libraries and institutions and their users to microforms as an alternative to bulky expensive print materials. By the 1970’s libraries and institutions began to use microfilming as an alternative to bulky expensive print materials and the improved technology also increased computer output microform applications. The application of microfilming was thus realised by the libraries and librarian in the late 19th and early 20th century.

Microfilm Formats

A number of different microfilm formats have come into being owing to different user needs and applications. The most common forms of microfilm are:

Roll Microfilm: It is the commonly used microform with images arranged in a linear array. The typical film widths are 16mm, 35mm, 70mm and 105mm. The film lengths range from 50 feet to more than 1000 feet, with 100 and 215 feet being the most widely encountered in library applications. The film may be unperforated or perforated along one or both edges, Non-perforated film is usually preferred for maximum utilisation of film area. A 100 feet reel of 16mm microfilm can store miniaturised images of 2,500 to 3,000 pages reduced by a ratio of 24:1. However, the recording/storage capacity of microfilm depends on the reduction ratio.

To facilitate easy handing the roll microfilms are loaded into self threading *catridges* and cassettes. The *catridge* is a closed container of 16mm film designed to load and unload in a reader or projector. It protects the film from fingerprints, dust and other possible damages. Whereas, cassettes are double spindled cartridges providing additional protection for handing the film, as in cassette it is not necessary to touch the film for rewind.

Unitized Microfilm: To facilitate ease of search of a document, the roll microfilms are many times split into small lengths, each of which becomes a unit, referred to as unitized microfilm. A strip of unitised microfilm usually contains 6 frames, the first frame being the title of the document. Thus, each strip of microfilm contains usually 10 pages of information.

Aperture Card: The aperture card is an opaque card of approximately 7"x 3" with an aperture at the right hand side for mounting a single 35mm frame. The card contains information about the image and is easy for storage and retrieval. Aperture cards are found heavily used in art, engineering and geography related departments. A sample of the aperture card is given below.

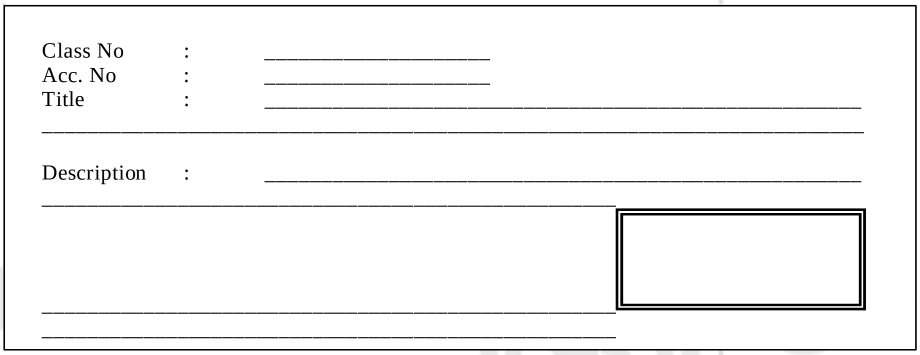

Fig. 15.1: Aperture Card

Jacket: The microfilm jackets are made of polyester sheets or keeping strips of microfilm, both 16mm. the size of the jackets are usually 105x148mm and the channels for keeping strips are made according to the need of keeping 16mm/35mm strips. The top of the jackets usually contains a heading area that gives information about the microfilm strips present in the jacket. Jacket are commonly found in large hospital and law firms for keeping individual cases, which are active and need to be updated.

Microfiche: Microfiche was first developed by Dr. J. Geobel of the Netherlands. It consists of a number of rows of reduced images of documents produced on a transparent sheet of film, using a special "Step and Repeat" camera. The microfiche is produced in variety of sizes, but the most common size is 105mm×148mm. the top of the fiche contains an eye legible image giving information about the content of the microfiche. Depending on the ratio of reduction, the microfiche vary in the number of frames. Ninety-eight frame microfiche are commonly found in libraries. Microfiche can be easily transported from one place to other and hence, now-a-days a number of publications are available in microfiche. Some commonly found reduction ratio and frame numbers per microfiche are given in table 15.1.

Table 1: Microfiche Reduction Ratio and Number of Frames

Reduction Times	No. of Rows	No. of Column	No. of Frames
18	05	12	60
24	07	14	98
42	13	16	208
48	15	18	270

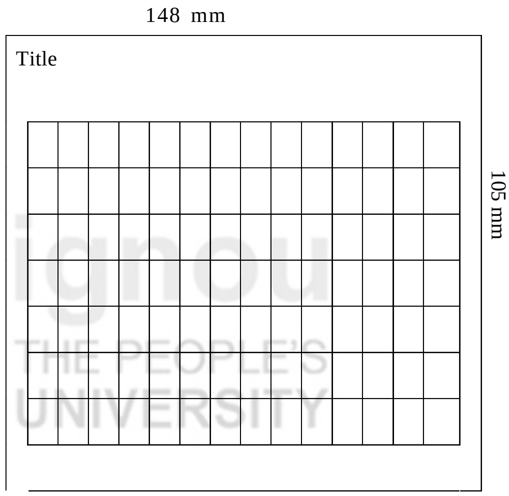

Fig. 15.2: Microfiche (98 frame)

Ultrafiche: It is fiche of the same size to that of microfiche, having significantly high number of frames. The original is reduced over 100 times to prepare ultrafiche. This is possible due to the availability of high quality film. Ultrafiche requires extreme care, as a small scratch can impair legibility of a number of frame due to its high frame capacity.

Micro-Card: This is an opaque card of 3"x5" size containing a number of rows of reduced images produced by photographic process.

Micro-Print: It is larger than micro-card having a size of 6"x 9". The images are printed by photolithography. Each card consists of 100 images with an eye-legible bibliographic reference along one side.

Microlex: The microlex cards are approximately of 6.5"x8.5" size and contain 200 pages on one side. The image are produced by photographic methods.

Self Check Exercise

1) Mention the common forms of microfilms in use with two sentences on each.

Note: i) Write your answer in the space given below.

ii) Check your answer with the answers given at the end of this Unit.

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Stability of Recorded Information

The information content of the microform is of utmost importance. American National Standard has identified 3 levels of stability of processed microforms/microfilms:

i) Archival Microfilms: Retaining original information-bearing characteristics indefinitely. They are suitable for preservation of information of permanent value.

ii) Medium-term Microfilms: Retaining the original information bearing characteristics for at least 10 years.

iii) Long-term Microfilms: Retaining the original information bearing characteristics for at least 100 years.

In fact, the level of stability associated with any given microform depends on various factors including the composition of photographic emulsion and microfilm base materials, the manner in which the film is developed and the conditions under which it is stored or used.

Depending on the type of emulsion type used the microfilms are of three types - Silver, Diazo and Vesicular Microfilm.

Silver-Halide Microfilm: It is high performance, fine grain photographic medium suitable for archival purpose. The photosensitive emulsion in the film contains silver halide crystals suspended in a gelatin matrix coated into a substrate composed of cellulose triacetate or polyester terapthalate or cellulose nitrate. Cellulose nitrate base microfilms are not common these days. The stability of processed silver halide microfilm may be affected by substrate characteristics as well as the characteristics of the chemical compounds used in the photographic recording process. All acetate and polyester based films are subject to dimensional changes associated with aging and environmental effects. In most cases, such changes are slight and have no impact on the utility of recorded information. Polyester based films may shrink by 0.03 percent over a period of 10 years, while acetate based films may shrink as much as 0.7 percent during the same period. Microfilm size also expand due to increase in temperature, the acetate or polyester based material may adhere to the emulsion layer of the portion of film wound immediately below it to separate the silver halide emulsion from its base, causing loss of recorded information. This problem is typically attributable to incomplete drying of processed film, prior to winding onto reels, high humidity in storage area, etc. For nitrate based films the stability is relatively less due to inflammable nature of the film. They are highly combustible and can ignite spontaneously at about 40 degree Celsius.

Diazo Microfilm: The diazo microfilms are usually used for duplicate or working copies. The emulsion used here is of diazonium salt. Usually the diazo films have a medium term stability with a usable life expectancy of at least 10 years. The diazo microforms may fade when exposed to light, including light encountered during usage. In diazo films the images are embedded in the base material as contrast to silver halide films, where the image is fixed to the base. Due to this the diazo films are less vulnerable to abrasion.

Vesicular Microfilm: the vesicular process is a system of dry photography involving exposure to ultraviolet light and development by heat. The vesicular microfilm consists of a light sensitive emulsion and when suspended is hard, but when exposed to ultraviolet radiation, pressure pockets/vesicules are formed as images. The images are then passed through heat for development and withdrawal of heat fixes the image and the film becomes hard. Stability of recorded information is similar to that of diazo films, but certain vesicular microfilms having long term storage potential may remain stable for 100 years when stored under controlled conditions. At high temperature, the vesicles collapse leading to loss of image. During storage, certain vesicular films release hydrogen chloride gas, which in the air produces hydrochloric acid. This has a corrosive effect on metal enclosures, film cans and cabinets. However, this corrosive effect has no harmful impact on the microfilm itself.

Self Check Exercise

1) Substrate characteristics can affect the stability of recorded information in microfilms. Comment.

Note: i) Write your answer in the space given below.

ii) Check your answer with the answers given at the end of this Unit.

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Advantages and Disadvantages of Microfilming

Advantages of Microfilming

i) **Life Expectancy**

The major advantage of microfilming is its durability and long life expectancy. As mentioned earlier a microfilm can last up to 500 years. For this it should be manufactured and processed to international quality standards; and stored under optimum environmental conditions.

ii) **Storage Space Reduction**

Another major advantage is that microfilm is very compact. A huge amount of information can be stored on microfilm in a much smaller space as compared to paper etc. Records reduced to microfilm occupy as little as 2% of space required for the original paper documents. Microfilming can help in realising a space saving of upto 98%.

iii) **Security of Information**

Microfilm helps to ensure security of vital or archival information. Once filmed it is not possible to manipulate the image on microfilm. To ensure preservation a separate security or master copy of the microfilm, on a polyester film base should be kept under strict security and environmental conditions. Moreover, retention of a master film copy acts as on backup ensuring that any tampering will be detected. This helps in maintaining the authenticity and integrity of the information.

iv) **Economical**

Another advantage of microfilming is savings in storage costs or records. The storage equipment requirements are less and thus economical. Microfilming also provides increased flexibility, and productivity in information management. Microfilms can be economically created, duplicated and distributed.

v) **Time-tested Method**

Microfilming has a well-proven history, as library material has been reproduced in microformats since 1930s. Longevity of the microfilmed information has been tested and problems, if any, with the technology have largely been ironed out.

vi) **Reducing Stress on Original Documents**

Creation of a microfilm copy provides an added benefit of reducing stress on the original material. Microfilming also helps to ensure the possibility of providing wider access by making duplicate copies for a range of users around the world. As once a record is microfilmed, it is easy and relatively inexpensive to make multiple copies of it by duplicating the film.

vii) **Option to Digitise**

Microfilm can also be digitised if good quality film has been used. Various reasons for microfilming of records exist.

Various reasons for microfilming of records exist. These factors can be used as measure to decide whether a particular record should be microfilmed or not. These factors are:

i) **Condition of the Document**

Condition of the document is the major determinant for its being microfilmed. The degree of deterioration or fragility should be considered. Records that are deteriorating, brittle, worn or water damaged can be microfilmed in order to preserve the information contained in them. Records that are beginning to show signs of disintegration can also be microfilmed to save them from further damage.

ii) **Rarity**

Microfilm is most useful for documents and records like manuscripts etc. which are rare and of intrinsic value. The degree of rarity is another factor that helps in microfilming whether a document should be microfilmed or not.

iii) **Frequency of use**

The level of usage of a record is another criterion to be considered for microfilming. Records that are used frequently are perfect candidates for microfilming, as microfilming will save on wear and tear of the originals that can be retained for preservation and enable production of multiple copies for use.

iv) **Volume of Space Occupied**

The volume of space occupied by the original records should be considered. Due consideration should be given to the fact that microfilming a large volume of record does save storage space but might cost more than what is required to film a small volume of record.

v) **Value**

Monetary, aesthetic and historical value attached to a record makes it more eligible to be microfilmed. All archival records valuable, however, some records like records of on going legal value, of high replacement cost or records that are irreplaceable, if lost, have more impact than others. Therefore, they should be microfilmed in order to securely preserve them.

vi) **Maintaining the integrity**

Microfilming safe grounds a document against manipulation. Therefore, if a document is of intrinsic value and its authenticity and integrity needs to be preserved it can be considered for microfilming.

All these factors should be considered before microfilming a document.

Disadvantages of Microfilming

i) **User Resistance:** The microfilmed records require a machine to be read. Often the microfilm reading machines in Libraries are of poor quality and not designed for human comfort. Usually, the users have to access the film manually by locating the film, loading it onto a machine and then spooling through dozens of microfilms to locate the one that is required or is of use. Therefore, some users are reluctant to use microfilm.

ii) **Cost:** Microfilming can be expensive. Costs incurred on microfilming include costs of equipment, cost of preparing documents for microfilming, and cost of producing indices and retrieval aids like microfilm readers etc.

iii) **Technical Limitations:** Microfilming has certain technical limitations also. If a document is faded, badly stained the microfilm image of that document will also be of poor quality. Moreover, the production of high quality archival microfilm is based on knowledge of technical requirements and imposition of rigid quality controls. Without this, even original documents in excellent condition can result in poor quality microfilms. Image quality can be determined only after filming is complete. Bad pages, if any, must be re-filmed and spliced in.

iv) **Damage to the Original:** The archival records being fragile may get damaged during the filming process. The microfilming process involves opening a bound volume and exposing the contents and filming it. This might crack or break binding and pages etc. if the original records are to be destroyed after filming this is not problematic. However, if the originals are to be retained, this is a drawback of the microfilming process.

The microfilms are used mainly for preservation purpose, but their use can't be restricted and, hence, there should always be at least two copies of the same document, one master and the other working copy. Master copies are seldom used or referred to. They are only used for duplicating purpose. Master copies should always be prepared by using silver halide microfilm, which has permanent life.

Self Check Exercise

2) What are the factors that should be considered while selecting material for microfilming?

Note: i) Write your answer in the space given below.

ii) Check your answer with the answers given at the end of this Unit.

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15.3 BASIC CONCEPT OF DIGITISATION

Digitisation may be defined as the process of converting information into a digital format. In this format, information is organised into discrete units of data (called bits) that can be separately addressed (usually in multiple-bit groups called bytes). This is the binary data that computers and many devices with computing capacity (such as digital cameras and digital hearing aids) can process. Also referred to as image capture, is the process of creating a digital representation or image of an original through scanning or digital photography. Digitization is a precondition for electronic storage, such as magnetic storage, storage on optical disk, and character recognition (eg, ICR, OCR).

By mid 1990's with the emergence of new technologies the shortcomings of microfilm as a medium of preservation become apparent. One of the earliest attempts at digitisation was undertaken at the British Library. The Electronic *Beowulf* project began as part of the British Library's "Initiatives for Access" programme to make its collections more available to the public through new technologies. The project produced the digital version of the unique late tenth-century manuscript of the Anglo-Saxon poem Beowulf, which was damaged by fire in 1731. High definition digital images of the valuable manuscript were made available to the scholars thus enabling new ways of accessing examining and researching the rare manuscript. This was made available in a CD-ROM which included "an electronically enhanced visual version of the text of the manuscript itself, important 18^{th}-century transcripts of the damaged original, a translation, a glossary, a textual commentary, references to relevant archaeological material and other editorial matter" (Feather, 1996). This attempt proved "the unlimited potential of digitisation as a tool for the preservation of texts and furtherance of scholarship" (Feather, 1996). The Electronic *Beowulf* project has till now assembled a huge library of digital images of the *Beowulf* manuscript and related manuscripts and inaccessible printed texts. Under the *Initiatives for Access* programme one project DAMP i.e., the Digitisation of Ageing Microfilm Project, highlighted the potential shift from microfilming to digitisation. The microfilms of Burney collection of Early English Newspapers, a national collection of 18^{th} Century London newspaper titles, were in a deteriorated state and the originals were too fragile to be re-filmed. DAMP aimed to digitise this collection by converting one surrogate form into another thus widening the access to content and making the texts available in a more convenient and machine-readable form. The British Library also started the International Dunhuang Project in 1993 to promote the study and preservation of manuscripts and printed documents from Dunhuang and other Central Asian sites through international co-operation. One of the pioneering project is Turning the Pages Programme (http://www.bl.uk/onlinegallery/ttp/ttpbooks.html). Under this programme a system is developed using high quality digitised images, interactive animation and touch screen technology to simulate the act of turning the pages of book. The user can virtually 'turn' the pages of manuscripts in a realistic way, zoom in on digitised images and read or listen to notes provided for each page. The British Library is now offering this service for institutions, museums and libraries